Predictive Regulated Search

Homework Assignment

Goal

Choose a problem in which a decision is obtained by search.

Build a search system that can use *predictions about its search space* and can use those predictions to decide *where computation should be spent.*

The final system should not simply search all alternatives uniformly. It should observe what happens during search, predict something useful about unresolved alternatives, and regulate subsequent search accordingly.

The choice of problem, representation, search algorithm, predictive model, training procedure, regulator, and experimental methodology is yours.

The purpose of the assignment is *not* to reproduce a specific algorithm from the lectures. The central question is

> Can prediction help the machine decide where further search is worth performing?

1. Search

Begin from a reasonable search procedure for your chosen problem.

Make explicit what, in your implementation, constitutes

state or symbolic search object, expansion, search horizon, active frontier.

You should be able to explain the distinction

> horizon $\neq$ active frontier.

The *horizon* is the boundary at which the current explicit search stops.

The *active frontier* is the part of that boundary still eligible to receive additional computation.

Explain what happens to alternatives that are no longer on the active frontier.

2. Introduce Prediction

Your system must contain at least one learned predictive component.

You must decide *what should be predicted.*

Possible choices include, but are not limited to:

- eventual value of a state;
- probability of success;
- solution quality;
- cost-to-go;
- preferred action;
- uncertainty of an estimate;
- expected error produced by stopping search;
- some other quantity that you can justify.

Denote your prediction generically by

$$\widehat{P}_\theta(s)$$

or, if it is action-dependent,

$$\widehat{P}_\theta(s, a).$$

You are not required to use a particular machine-learning model. A linear model may be enough. A neural model may be appropriate. The important issue is not model complexity but what the prediction means and how search uses it.

3. Where Do the Targets Come From?

Construct a way of obtaining targets for your predictor.

One possibility is to use a more expensive search as a teacher:

$$s \longrightarrow \text{deeper / more complete search} \longrightarrow P^*(s).$$

The cheaper predictor then learns

$$\widehat{P}_\theta(s) \approx P^*(s).$$

Exact solutions are not required if they are impractical. A stronger search, larger budget, simulation, or another independently justified procedure may provide the target.

Explain:

- how training examples are produced;
- what the target means;
- why that target should be useful to subsequent search.

Keep the computation used for training conceptually separate from the *online search budget* used in the final comparison.

4. Predictive Search

Use your prediction inside the search.

For example, prediction might determine frontier ordering, leaf evaluation, candidate ordering, probability assigned to actions, or another aspect of the search.

Construct a **predictive but non-regulated** version of the system.

This distinction is important:

prediction: what looks promising?

regulation: how much computation should it receive?

5. Regulate the Predictive Search

Now introduce a regulator.

The regulator must observe some property of the ongoing search and modify the allocation of computation.

Possible observables include

$$|\mathcal{F}_t|, \qquad \beta_t, \qquad H_t, \qquad \rho_t,$$

prediction confidence, disagreement between predictors, change of prediction with search depth, bound width, or another quantity appropriate to your problem.

The regulator may control candidate width, local depth, expansion priority, stopping criteria, temperature, or some other computational resource.

Schematically,

$$\mathcal{O}_t \longrightarrow \mathcal{R} \longrightarrow \mathcal{U}_t,$$

where $\mathcal{O}_t$ contains observations of the search and $\mathcal{U}_t$ determines how subsequent computation is allocated.

Explain precisely what this map means in your implementation.

6. Prediction Error and Need for Search

A prediction does not have to be correct. Its error may itself be useful.

Whenever possible, compare a cheaper prediction with the result produced by additional search. For example,

$$r(s) = P_{\text{search}}(s) - \widehat{P}_\theta(s).$$

Ask whether

$$|r(s)|$$

can identify states for which stopping search is dangerous.

You may attempt to learn this quantity as well,

$$\widehat{R}_\phi(s) \approx |r(s)|,$$

but a second learned model is not mandatory. Another justified estimate of uncertainty or need for search is acceptable.

A central question is

Can the system distinguish "looks good" from "needs more thinking"?

7. Probabilistic Choice

At some meaningful decision point, represent alternatives through a probability distribution

$$\pi(a \mid s).$$

You decide how this distribution is obtained. It may come from a learned model, predicted values, a softmax transformation, empirical frequencies, or another justified construction.

Explain what these probabilities mean in your system.

Some alternatives may receive additional explicit search while others do not. Identify the probability mass associated with the unresolved complement:

$$\rho_t = \sum_{a \in U_t} \pi(a \mid s),$$

or define an analogous quantity appropriate to your problem.

Explain what it means for this probability mass to remain unresolved.

8. Experiment

Compare at least the following three systems:

A. Baseline search
B. Predictive search
C. Predictive + regulated search

They should operate under approximately the same **online computational budget**.

Your experiment should distinguish

gain from prediction

from

gain from regulation.

Choose measurable criteria appropriate to your problem. Possible measures include

decision accuracy, regret, solution quality, nodes expanded,

effective depth, residual probability, prediction error.

Node count alone is not a measure of decision quality.

Whenever feasible, construct some independent way of determining a better or correct decision on at least a subset of the test cases.

A negative result is acceptable. If regulation does not help, explain why you think it does not help.

9. Questions Your Report Must Answer

Your report should make it possible to answer:

1. What exactly is being searched?
2. What is a state or symbolic alternative?
3. What is the search horizon?
4. What part of the horizon is the active frontier?
5. What did you choose to predict?
6. Where did the training targets come from?
7. How accurate is the predictor?
8. How is prediction used by search?
9. What additional information does the regulator observe?
10. What computational decision does the regulator make?
11. What does $\pi(a \mid s)$ mean in your implementation?
12. What constitutes unresolved probability mass?
13. What happens when the predictor is wrong?
14. Under equal online resources, what is gained by prediction?

15. What additional gain, if any, comes from regulation?
16. Did you find cases in which further search is valuable even though the predicted value itself looks uninteresting?

The final question is particularly important.

10. Deliverables

Submit:

- an executable notebook or program containing the complete experiment;
- the code used to construct or obtain the predictive model;
- a short report explaining the design and results;
- sufficient data or generated instances to reproduce the main experiment;
- a short **AI decision record** containing 3–5 AI interactions that materially affected your solution.

For each reported AI interaction, state briefly:

- what you asked;
- what the AI proposed;
- whether you accepted, rejected, or modified the proposal;
- why.

A complete transcript is not required.

Use of AI

Use of ChatGPT or other AI systems is explicitly allowed and encouraged.

You may use AI to design the predictor, write code, select models, debug the implementation, generate experiments, derive formulas, analyze results, and help write the report.

There is no bonus for avoiding AI.

However, you are responsible for the system that you submit.

You should understand not only what the program does, but why its predictive and regulatory components affect the search in the way that they do.

Subsequent Verification

Part of the assessment of this homework will occur later.

You may be given a modification of your own system and asked to reason about its consequences before executing it.

For example, you may be asked what happens if:

- the predictor becomes less accurate;
- the probability distribution is flattened;
- the search budget is reduced;
- one source of uncertainty is removed;
- the regulator is disabled;
- prediction is retained but computation is allocated uniformly;
- residual probability is forced to remain larger;
- the training distribution changes.

You may also be asked to make a small modification to your implementation.

The assessment therefore concerns not only the quality of the system you produced, but how well you understand the relation

$$\boxed{\text{prediction} \longrightarrow \text{search} \longrightarrow \text{regulation} \longrightarrow \text{decision}.}$$